SHIHAO ZHUO

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EDUCATION

University of Birmingham, Artificial Intelligence and Machine Learning, *MSc* 2026.09 - present

Ningbo University, Computer Science and Technology, *BEng* 2018.09 - 2024.06

- The six-year span includes two years of compulsory military service (2019.09–2021.09).
- Third-Class University Scholarship; Outstanding League Member; Outstanding Student Cadre (university level); Outstanding League Branch Secretary.

PUBLICATIONS

- **RMD^2: One Geometry, Two Roles in Data Curation and Generation Guidance Under review, VLDB 2027**
  Co-first author. A **low-cost curation operator** for large data pipelines. Class statistics – an MRCD robust covariance and a regularised precision matrix – are fitted **once** from a small trusted reference set in a frozen embedding space; scoring a candidate is then one encoder pass and one quadratic form, $O(NDk)$, with **no task-proxy model trained anywhere in the loop**. An inter-class Mahalanobis margin, rather than an absolute distance, decides whether a sample agrees with its assigned label. The fitted statistics are cached and reused across pool refreshes, downstream tasks, and even the switch from selecting data to generating it.

  Cost: 69s / 47s end-to-end on CIFAR-10N / CLINC150, the fastest of eight compared methods; 793MB peak memory, 19.7× below the runner-up SimiFeat; proxy-based attribution (IF / TRAK / LESS) costs 2–3 orders of magnitude more time because it trains a network first. **Quality is not traded away**: 98.8% / 99.9% detection AUROC, 0.56% / 0.02% residual noise in the selected set, 11.2–15.4 downstream accuracy points over reference-only training. The same cached object guides generation directly – conditional-GAN FID -11.0% and KID -18% (paired test, $p < 0.05$), and as a DPO preference signal it raises augmentation accuracy by up to 1.22 points – with no second quality model to train.

RESEARCH AND PROFESSIONAL EXPERIENCE

HKUST (Guangzhou), Research Assistant 2026.03-2026.07

- Responsible for the method design, experiments and paper writing of the RMD^2 framework, submitted to VLDB 2027 (see Publications).
- Key design: replacing the absolute Mahalanobis distance with an inter-class margin as the label-consistency criterion raised detection AUROC on CIFAR-10N from 91.98 to 98.84, so that hard but correctly labelled samples are no longer discarded.
- Built a two-modality pipeline (DINOv2-ViT-S/14, BGE-large-en-v1.5, ResNet-9, BERT-base, Qwen2.5-1.5B with LoRA/DPO) and benchmarked it on 4×RTX A5000 against IF, TRAK, LESS, DataInf, SimiFeat, LAVA, CLIP and ImageReward under identical data splits.

Shanghai Jiao Tong University, Research Assistant 2024.06-2024.12

- Independently built and maintained the research group website on XAMPP 8.0 and WordPress 5.8 (`https://aipsybrain.com/`).
- Independently developed a Python GUI application for cognitive assessment tasks, supporting memory and cognition scales; deployed for hospital-side data collection.

Anlai Scientific Instruments (Ningbo) Co., Ltd., Intern Assistant 2024.07-2024.10

- Contributed to a mouse scratching-behaviour analysis project, applying YOLOv8 for detection data processing and model tuning.
- Audited annotation quality and tuned algorithm parameters to keep training data accurate and model performance stable.

Ningbo Fanxing Consulting Co., Ltd., Development Engineer (Intern) 2023.04-2023.07

- Developed a college-application recommendation system as a WeChat Mini Program, using WeChat Cloud's built-in database to simplify storage; the work obtained one software copyright.

PLA Navy, Northern Theater Command, Soldier (compulsory military service) 2019.09-2021.09

Research Projects

- **Multi-Center MRI Image Classification Based on Deep Learning** 2024.01-2024.05
  To address the loss of generalisation caused by multi-center data heterogeneity, diffusion models and domain-transfer techniques align unseen-center data with known-center data, improving cross-center transferability of the classifier; the method was filed as a patent.
- **Accurate Rib Fracture Diagnosis with Lightweight Deep Models** 2022.03-2023.03
  A lightweight network for chest CT images that localises and classifies rib fracture regions and generates diagnostic reports, and raises an alert when the radiologist's reading disagrees with the model prediction, reducing missed and incorrect diagnoses.

Patents and Software Copyrights

- **Multi-Center MRI Image Classification Method**, Invention Patent 2024.08
  Granted, CN117173489B; **first inventor**. A vertical loss separates content features from style features so that the classifier attends to content, mitigating cross-center imaging heterogeneity and improving generalisation.
- **A Sequential Multi-Classification Method for sMRI Images**, Invention Patent 2024.08
  Granted, CN116740463B; fifth inventor. A ranking model keeps distances between class feature vectors consistent with their distances along the actual pathological progression, giving the features an ordinal structure and improving classification.
- **sMRI Classification with a High-Resolution Complementary-Attention UNet**, Invention Patent 2022.07
  Granted, CN114332535B; fifth inventor. Feature-fusion and classification branches inserted into UNet supply semantic information to the encoder and detail to the decoder, and fuse multi-semantic feature maps to improve classification accuracy.
- **Fanxing Sanyi Information Consulting Mini Program**, Software Copyright 2023.05
  A consulting platform for Zhejiang's "Trinity" admission scheme: it recommends eligible universities from a candidate's academic exam results and provides per-university strategies and admission records.

Technical Skills

- **Programming**: Python (proficient), C/C++, SQL, JavaScript/HTML/CSS, LaTeX
- **Machine learning**: PyTorch, HuggingFace Transformers / PEFT (LoRA) / TRL (DPO), scikit-learn, NumPy/SciPy, YOLO, multi-GPU training
- **Tools and platforms**: Linux, macOS, Git, MongoDB, openGauss, WordPress
- **Languages**: Chinese (native), English (MSc taught in English; research written in English)